

Bioremediation Rubric

	3	4	5	6	7	8
Asking questions	Questions are relevant to specific topic.	Questions are relevant to a specific topic and are testable or researchable.	Questions are valid, testable or researchable, and based on patterns or observations.	Questions are valid, focused, testable or researchable, based on patterns/observations, current research, and/or a specific model or theory.	Questions are valid, precise, testable or researchable, and based on patterns/observations, specific evidence from current research, and/or a specific model or theory.	Questions are valid, precise, testable or researchable, based on patterns/observations, specific evidence from current research and/or a specific model or theory, and push standard thinking on a given topic or in a particular discipline.
Hypothesizing	Provides a prediction or early hypothesis with some relationship to the question under investigation.	Articulates a relevant prediction of the expected results with relation to the question under investigation, but variables are unclearly stated.	Constructs a testable hypothesis about the investigated question, with a basic description of the variables ("if.. then..."). Hypothesis relates to observation, research, or scientific principle.	Constructs a clear, testable hypothesis about the investigated question, with an accurate description of the variables ("if... then..."). Hypothesis is based on observation, research, scientific principle, model, or theory.	Constructs a precise, testable hypothesis about the investigated question, with an accurate explanation of the relationship between variables ("if... then... because..."). Hypothesis is based on observation, research, scientific principle, model, or theory.	Constructs a precise, testable, and insightful hypothesis about the investigated question, with accurate and thorough explanation of the relationship between variables ("if... then... because"). Hypothesis is based on observation, research, scientific principle, model, or theory.
Designing Processes and Procedures	Identifies a starting point to address a problem/prompt and creates useful follow-up steps, though steps may be out of order or may not fully address the prompt.	Identifies a starting point to address a problem/prompt and organizes useful follow-up steps in a logical, sequential order. May not fully address the prompt.	Creates an orderly action plan that addresses all aspects of a problem/prompt with some inefficiencies.	Creates a clear and orderly action plan that is mostly replicable and addresses all aspects of a problem/prompt in an efficient way. Includes a rationale for some steps or aspects of the plan.	Creates a clear, detailed action plan that is fully replicable and addresses all aspects of a problem/prompt in an efficient way. Includes a rationale for several steps or aspects of the plan. Where applicable, includes some alternate or contingency plans.	Creates a clear, detailed, fully replicable action plan to address a problem/prompt as efficiently as possible. Includes a rationale for the complete plan. Where applicable, acknowledges potential weaknesses or limitations of the plan and includes thorough alternate or contingency plans.
Identifying Patterns and Relationships	Organizes information into mostly useful structures. Identifies patterns with some inaccuracies.	Organizes information into mostly useful structures and accurately identifies patterns.	Organizes information into useful structures. Accurately identifies patterns and some relationships among patterns.	Organizes information into useful structures. Accurately identifies significant/relevant patterns and relationships among patterns.	Organizes information into useful structures. Identifies and explains significant/relevant patterns and the relationships among patterns. Makes note of examples and data that do not fit the pattern(s) or relationship(s).	Organizes information into useful structures. Identifies and explains significant/relevant patterns and relationships among patterns. Identifies and explains examples and data that do not fit the pattern(s) or relationship(s).

Interpreting Data/Info	Provides some partial or general analysis of data/information. May contain errors.	Provides a reasonable interpretation of data/information. May contain minor errors.	Provides an accurate interpretation of data/information. Applies some analytic strategies (e.g., sorting, compare/contrast) or concepts (e.g., mean, mode) to characterize the data/information.	Provides an accurate interpretation of data/information. Applies appropriate analytic strategies or concepts to characterize the data/information. Considers the context from which the data/information arose.	Provides a thorough, accurate interpretation of data/information. Applies multiple analytic strategies or concepts to characterize the data/information. Considers the context from which the data/information arose. Recognizes gaps or outliers in the data/information.	Provides a thorough, accurate interpretation of data/information. Applies multiple analytic strategies/concepts and determines which strategy/concept is best for the purpose of the analysis. Considers the context from which the data/information arose. Explains gaps or outliers in the data/information.
Making Connections & Inferences	Makes inferences that are based on evidence but may be partially formed with gaps in explaining the connection of a specific example to the larger idea.	Makes relevant inferences based on evidence and attempts to identify the larger significance of the inference. Connections between a specific example and the larger idea are clear and appropriate.	Makes clear and relevant inferences based on evidence and partially explains the larger significance of the inference. Connections to the larger idea are made through multiple examples but may have some gaps in explanation or may not be fully developed.	Makes clear, relevant, thoughtful inferences and explains their larger significance. Where applicable, identifies limitations of inferences based on gaps in evidence. Connections to the larger idea are clearly made through multiple examples.	Makes clear, highly relevant and thoughtful inferences and thoroughly explains their larger significance. Where applicable, identifies limitations of inferences based on gaps in evidence. Uses inferences as the basis for predictions or broader generalizations. Connections to the larger idea are clearly made through multiple examples, including attempts at non- or counter-examples.	Makes clear, highly relevant, insightful inferences and thoroughly explains their larger significance with sophisticated insight or originality of interpretation. Where applicable, identifies limitations of inferences based on gaps in evidence. Uses inferences as the basis for predictions or broader generalizations. Connections to the larger idea are clearly made through multiple examples, including non- or counter-examples.
Explanation of Evidence	Provides mostly relevant analysis that partially explains how selected evidence supports claims or statements; may still contain some repeating, rewording, and/or summarizing of evidence.	Provides relevant analysis that explains how the selected evidence supports claims or statements; analysis stays rooted in the evidence but at times may be vague, illogical, or overly general.	Provides clear analysis that accurately explains how the selected evidence supports claims or statements.	Provides insightful and clear analysis that thoroughly and accurately explains how the evidence supports claims or statements; where applicable, analysis acknowledges some weakness(es) or gaps in the evidence.	Provides insightful, clear, compelling analysis that thoroughly and accurately explains how the evidence supports claims or statements; where applicable, analysis addresses weakness(es) or gaps the evidence.	Provides insightful, compelling analysis that thoroughly, accurately, and concisely explains how the evidence supports claims or statements; where applicable, analysis clearly addresses weakness(es) or gaps in the evidence; analysis is elegant in its precision and/or sophistication and originality.

Organization (Transitions, Cohesion, Structure)	Paragraphs or sections are connected around a main idea but do not clearly build on one another. Transition words/phrases are present but are often formulaic.	Paragraphs and/or sections are connected and sequenced to support understanding of ideas. Transitions are varied and are mostly appropriate and effectively used.	Paragraphs and/or sections are connected and logically build upon one another to deepen understanding of ideas and clarify relationships among ideas. Transitions are varied and appropriately and effectively used.	Paragraphs and/or sections are connected and clearly and logically build upon one another to deepen understanding of complex ideas and to clarify relationships among those ideas. Transitions are varied and appropriately and effectively used. Sequencing of paragraphs and use of transitions help build cohesion.	Sequencing of paragraphs and/or sections creates a coherent whole that deepens understanding of the content and builds toward a particular outcome. Transitions are appropriate, effective, and varied in their structure and location.	Sequencing of paragraphs or sections creates a coherent whole that deepens understanding of the content and clearly guides the reader toward a particular outcome. Transitions are appropriate, effective, and varied in their structure and location. Sequencing of ideas and transitions is seamless and fluid, and enhances the purpose of the writing.
Discussion / Contribution	Expresses some original ideas and makes some connection to the ideas of others. Questions and responses demonstrate some use of high-level questioning strategies and/or higher order thinking.	Expresses original ideas clearly and connects to the ideas of others. Questions and responses are mostly high level. Attempts to move discussion forward by asking and responding to questions.	Expresses original ideas clearly and persuasively; connects to the ideas of others and builds new pathways of discussion. Attempts to deepen discussion by asking connecting questions or building on the responses of others.	Expresses original ideas clearly and persuasively. Builds new pathways of discussion that are clearly connected to the ideas of others. Propels conversations by relating to broader themes.	Expresses original ideas clearly and persuasively. Builds new pathways of discussion that are clearly connected to the ideas of others. Uses questions and summarization to preserve focus. Propels conversations by relating to broader themes.	Expresses original ideas clearly and persuasively. Builds new pathways of discussion that are clearly connected to the ideas of others. Uses questions and summarization to preserve focus. Propels conversations by relating to broader themes, probing reasoning and evidence and/or promoting divergent and creative perspectives.
Preparation	Prepares general notes with some specific connections to highlight key talking points.	Prepares specific notes with connections to highlight key areas. Explicitly draws on those notes to probe and reflect on ideas under discussion.	Comes to discussions having read & researched material for teacher-provided questions; explicitly draws on texts & research to stimulate a thoughtful, well-reasoned exchange of ideas.	Come to discussions with responses and evidence generated with peers through studying, research, or inquiry; explicitly draws on texts and research to stimulate a thoughtful, well-reasoned exchange of ideas.	Come to discussions with self-generated questions from studying, research, and/or inquiry. Attempts to move discussion along by using those questions at appropriate moments.	Come to discussions with self-generated, sequenced, specific questions and evidence to move the discussion along purposefully (i.e., toward an intended outcome or desired understanding).